Numerical Analysis

Chapter 6: Numerical Differentiation and Integration

Department of Mathematics & Statistics
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Textbook: Classical and Modern Numerical Analysis:
Theory, Methods and Practice
A. S. Ackleh, E. J. Allen, R. B. Kearfott and P. Seshaiyer
CRC Press, Boca Raton, FL, 2010

Outline

1. Numerical differentiation
2. Numerical integration/quadrature
3. Other topics

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2 Numerical integration/quadrature

3 Other topics

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Derivation with Taylor's theorem

Recall

Theorem

Suppose that $f(x)$ is a $(n+1)^{th}$ differentiable function on $[a,b]$ and $x_0 \in [a,b]$. Then for any $x \in [a,b]$, we have the following Taylor's expansion of $f(x)$ at x_0:

$$f(x) = P_n(x) + R_n(x),$$

where

$$\begin{aligned} P_n(x) &= \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x-x_0)^k \\ &= f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!} f''(x_0)(x-x_0)^2 + \cdots \\ &\quad + \frac{1}{n!} f^{(n)}(x_0)(x-x_0)^n, \end{aligned}$$

Derivation with Taylor's theorem

Theorem (Continued)

$$R_n = \frac{1}{(n+1)!} f^{(n+1)}(\xi)(x - x_0)^{n+1}$$

for some $\xi \in [x_0, x]$ *(Lagrange form of the remainder)* ,

or

$$R_n = \frac{1}{n!} \int_{x_0}^{x} f^{(n+1)}(s)(x - s)^n \, ds$$

(Integral form of the remainder) .

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- Pick $n = 3$ in the Taylor's expansion:

$$\begin{aligned} f(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2 \\ &\quad + \frac{1}{6} f'''(x_0)(x - x_0)^3 + \frac{1}{24} f^{(4)}(\xi)(x - x_0)^4. \end{aligned}$$

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- Replace x by $x + h$ and x_0 by x:

$$f(x + h) = f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 + \frac{1}{6}f'''(x)h^3 + O\left(h^4\right).$$

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- We first consider the discretization of the first derivative $f'(x)$. Then

$$\begin{aligned} f'(x) &= \frac{f(x + h) - f(x)}{h} - \frac{1}{2}f''(x)h - \frac{1}{6}f'''(x)h^2 - O\left(h^3\right) \\ &= \frac{f(x + h) - f(x)}{h} + O\left(h\right). \end{aligned}$$

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- Then

$$\begin{aligned} f'(t_j) &= \frac{f(t_j + h) - f(t_j)}{h} + O(h) \\ &= \frac{f(t_{j+1}) - f(t_j)}{h} + O(h) \\ &\approx \frac{f_{j+1} - f_j}{h},\ j = 0, 1, \cdots, J-1. \end{aligned}$$

Here f_j is the approximation of $f(t_j)$. This is called forward difference.

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- Recall the Taylor's expansion with $n = 3$:

$$\begin{aligned} f(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2 \\ &\quad + \frac{1}{6} f'''(x_0)(x - x_0)^3 + \frac{1}{24} f^{(4)}(\xi)(x - x_0)^4. \end{aligned}$$

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- Replace x by $x - h$ and x_0 by x:

$$f(x - h) = f(x) - f'(x)h + \frac{1}{2}f''(x)h^2 - \frac{1}{6}f'''(x)h^3 + O\left(h^4\right).$$

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$$\begin{aligned} f'(x) &= \frac{f(x) - f(x - h)}{h} + \frac{1}{2}f''(x)h - \frac{1}{6}f'''(x)h^2 + O\left(h^3\right) \\ &= \frac{f(x) - f(x - h)}{h} + O\left(h\right). \end{aligned}$$

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$$\begin{aligned} f'(t_j) &= \frac{f(t_j) - f(t_j - h)}{h} + O(h) \\ &= \frac{f(t_j) - f(t_{j-1})}{h} + O(h) \\ &\approx \frac{f_j - f_{j-1}}{h}, \ j = 1, \cdots, J. \end{aligned}$$

This is called backward difference.

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- Recall

$$f(x+h) = f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 + \frac{1}{6}f'''(x)h^3 + O\left(h^4\right),$$

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$$f(x-h) = f(x) - f'(x)h + \frac{1}{2}f''(x)h^2 - \frac{1}{6}f'''(x)h^3 + O\left(h^4\right).$$

- Subtract the second equation from the first one:

$$f(x+h) - f(x-h) = 2f'(x)h + \frac{1}{3}f'''(x)h^3 + O\left(h^4\right).$$

Derivation with Taylor's theorem

- Then

$$\begin{aligned} f'(x) &= \frac{f(x+h)-f(x-h)}{2h} - \frac{1}{12}f'''(x)h^2 + O\left(h^3\right) \\ &= \frac{f(x+h)-f(x-h)}{2h} + O\left(h^2\right). \end{aligned}$$

This is second order!

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- Hence

$$\begin{aligned} f'(t_j) &= \frac{f(t_j+h)-f(t_j-h)}{2h} + O\left(h^2\right) \\ &= \frac{f(t_{j+1})-f(t_{j-1})}{2h} + O\left(h^2\right) \\ &\approx \frac{f_{j+1}-f_{j-1}}{2h},\ j=1,\cdots,J-1. \end{aligned}$$

This is called centered difference.

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- Pick $n = 4$ in the Taylor's expansion:

$$\begin{aligned} f(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 + \\ &\frac{1}{6}f'''(x_0)(x - x_0)^3 + \frac{1}{24}f^{(4)}(x_0)(x - x_0)^4 \\ &+ \frac{1}{120}f^{(5)}(\xi)(x - x_0)^5. \end{aligned}$$

Derivation with Taylor's theorem

- Replace x by $x+h$ and x_0 by x:

$$\begin{aligned} f(x+h) &= f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 + \frac{1}{6}f'''(x)h^3 \\ &\quad + \frac{1}{24}f^{(4)}(x)h^4 + O\left(h^5\right). \end{aligned}$$

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- Replace x by $x+h$ and x_0 by x:

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- Replace x by $x-h$ and x_0 by x:

$$\begin{aligned} f(x-h) &= f(x)-f'(x)h+\frac{1}{2}f''(x)h^2-\frac{1}{6}f'''(x)h^3 \\ &\quad +\frac{1}{24}f^{(4)}(x)h^4+O\left(h^5\right). \end{aligned}$$

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- Add the above two equations together:

$$f(x+h)+f(x-h)=2f(x)+f''(x)h^2+\frac{f^{(4)}(x)}{12}h^4+O\left(h^5\right) \quad .$$

Derivation with Taylor's theorem

- Then

$$\begin{aligned} f''(x) &= \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} - \frac{f^{(4)}(x)}{12}h^2 + O\left(h^3\right) \\ &= \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} + O\left(h^2\right). \end{aligned}$$

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- Forward difference for $f'(x_j) \approx \frac{f_{j+1}-f_j}{h}$.

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Derivation via Lagrange polynomial representation

Recall the following theorem from section 4.3:

Theorem

For any $n+1$ distinct real numbers $x_0, x_1, x_2, \cdots, x_n$ and for $n+1$ arbitrary numbers $y_0, y_1, y_2, \cdots, y_n$, let

$$L_k(x) = \prod_{i=0, i\neq k}^{n} \frac{x - x_i}{x_k - x_i}.$$

Then there exists a unique interpolating polynomial

$$p(x) = \sum_{k=0}^{n} y_k L_k(x)$$

of degree at most n such that $y_j = p(x_j)$ $(j = 0, 1, 2, \cdots, n)$.

Derivation via Lagrange polynomial representation

Recall the following definition and theorem from section 4.3:

Definition

$p(x) = \sum_{k=0}^{n} f(x_k)L_k(x)$ is called the Lagrange polynomial approximation/interpolation of a given function $f(x)$.

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Theorem (II)

If $x_0, x_1, x_2, \cdots, x_n$ are $n+1$ distinct points in $[a, b]$ and $f \in C^{n+1}[a, b]$, then for each $x \in [a, b]$, there exists a number $\xi = \xi(x) \in (a, b)$ such that

$$f(x) - p(x) = \frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!}$$

where $W(x) = \prod_{i=0}^{n}(x - x_i)$.

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- For $m = 1$: $f'(x) \approx p'(x) = \sum_{k=0}^{n} f(x_k) L'_k(x)$.
- And $f'(x_l) \approx p'(x_l) = \sum_{k=0}^{n} f(x_k) L'_k(x_l)$.

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- Basic idea: $f^{(m)}(x) \approx p^{(m)}(x)$ where $p(x) = \sum_{k=0}^{n} f(x_k) L_k(x)$.
- For $m = 1$: $f'(x) \approx p'(x) = \sum_{k=0}^{n} f(x_k) L_k'(x)$.
- And $f'(x_l) \approx p'(x_l) = \sum_{k=0}^{n} f(x_k) L_k'(x_l)$.
- Don't forget the remainder: $\frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!}$ where $W(x) = \prod_{i=0}^{n}(x - x_i)$.
- So

$$\begin{aligned} f(x) &= p(x) + \frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!} \\ &= \sum_{k=0}^{n} f(x_k) L_k(x) + \frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!}. \end{aligned}$$

Derivation via Lagrange polynomial representation

- $f'(x) = \sum_{k=0}^{n} f(x_k) L'_k(x) + W'(x) \frac{f^{(n+1)}(\xi(x))}{(n+1)!} + \frac{d}{dx} \frac{f^{(n+1)}(\xi(x))}{(n+1)!} W(x)$.

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- Recall $W(x) = \prod_{i=0}^{n}(x - x_i)$.
- So $W'(x) = \sum_{j=0}^{n} \prod_{i=0, i\neq j}^{n}(x - x_i) =$

$$\prod_{i=1}^{n}(x - x_i) + \prod_{i=0, i\neq 1}^{n}(x - x_i) + \cdots + \prod_{i=0}^{n-1}(x - x_i).$$

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- And $W'(x_l) = \prod_{i=0, i\neq l}^{n}(x_l - x_i)$ for any $0 \leq l \leq n$.

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- And $W'(x_l) = \prod_{i=0, i\neq l}^{n}(x_l - x_i)$ for any $0 \leq l \leq n$.
- Also, $W(x_l) = \prod_{i=0}^{n}(x_l - x_i) = 0$ for any $0 \leq l \leq n$.

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- And $W'(x_l) = \prod_{i=0, i\neq l}^{n}(x_l - x_i)$ for any $0 \leq l \leq n$.
- Also, $W(x_l) = \prod_{i=0}^{n}(x_l - x_i) = 0$ for any $0 \leq l \leq n$.
- Hence $f'(x_l) = \sum_{k=0}^{n} f(x_k)L'_k(x_l) + \prod_{i=0, i\neq l}^{n}(x_l - x_i)\frac{f^{(n+1)}(\xi(x_l))}{(n+1)!}$.

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Derivation via Lagrange polynomial representation

- Example 1: Pick $n = 2$, $x_1 = x_0 + h$, and $x_2 = x_1 + h$.
- Then $f'(x_l) = \sum_{k=0}^{2} f(x_k) L_k'(x_l) + \prod_{i=0, i\neq l}^{2} (x_l - x_i) \frac{f'''(\xi(x_l))}{6}$.
- Recall

$$L_k(x) = \prod_{i=0, i\neq k}^{n} \frac{x - x_i}{x_k - x_i}.$$

- Then

$$\begin{aligned} L_0(x) &= \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}, \\ L_1(x) &= \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}, \\ L_2(x) &= \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}. \end{aligned}$$

Derivation via Lagrange polynomial representation

- Hence

$$L_0'(x) = \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)},$$

Derivation via Lagrange polynomial representation

- Hence

$$
\begin{aligned}
L_0'(x) &= \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}, \\
L_1'(x) &= \frac{2x - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)},
\end{aligned}
$$

Derivation via Lagrange polynomial representation

- Hence

$$
\begin{aligned}
L_0'(x) &= \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}, \\
L_1'(x) &= \frac{2x - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)}, \\
L_2'(x) &= \frac{2x - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)}.
\end{aligned}
$$

Derivation via Lagrange polynomial representation

- Hence

$$
\begin{aligned}
L_0'(x) &= \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}, \\
L_1'(x) &= \frac{2x - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)}, \\
L_2'(x) &= \frac{2x - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)}.
\end{aligned}
$$

- And

$$
L_0'(x_l) = \frac{2x_l - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)},
$$

Derivation via Lagrange polynomial representation

- Hence

$$
\begin{aligned}
L_0'(x) &= \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}, \\
L_1'(x) &= \frac{2x - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)}, \\
L_2'(x) &= \frac{2x - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)}.
\end{aligned}
$$

- And

$$
\begin{aligned}
L_0'(x_l) &= \frac{2x_l - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}, \\
L_1'(x_l) &= \frac{2x_l - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)},
\end{aligned}
$$

Derivation via Lagrange polynomial representation

- Hence

$$
\begin{aligned}
L_0'(x) &= \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}, \\
L_1'(x) &= \frac{2x - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)}, \\
L_2'(x) &= \frac{2x - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)}.
\end{aligned}
$$

- And

$$
\begin{aligned}
L_0'(x_l) &= \frac{2x_l - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}, \\
L_1'(x_l) &= \frac{2x_l - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)}, \\
L_2'(x_l) &= \frac{2x_l - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)}.
\end{aligned}
$$

Derivation via Lagrange polynomial representation

- Hence for $l = 0, 1, 2$:

$$f'(x_l) = \sum_{k=0}^{2} f(x_k) L'_k(x_l) + \prod_{i=0, i\neq l}^{2} (x_l - x_i) \frac{f'''(\xi(x_l))}{6}$$

Derivation via Lagrange polynomial representation

- Hence for $l = 0, 1, 2$:

$$
\begin{aligned}
f'(x_l) &= \sum_{k=0}^{2} f(x_k) L_k'(x_l) + \prod_{i=0, i\neq l}^{2} (x_l - x_i) \frac{f'''(\xi(x_l))}{6} \\
&= f(x_0) \frac{2x_l - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)} + f(x_1) \frac{2x_l - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)} \\
&\quad + f(x_2) \frac{2x_l - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)} + \prod_{i=0, i\neq l}^{2} (x_l - x_i) \frac{f'''(\xi(x_l))}{6}.
\end{aligned}
$$

Derivation via Lagrange polynomial representation

- Pick $l = 1$:

$$\begin{aligned} f'(x_1) &= f(x_0)\frac{2x_1 - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)} + f(x_1)\frac{2x_1 - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)} \\ &\quad + f(x_2)\frac{2x_1 - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)} + \prod_{i=0, i\neq 1}^{2}(x_1 - x_i)\frac{f'''(\xi(x_1))}{6} \end{aligned}$$

Derivation via Lagrange polynomial representation

- Pick $l = 1$:

$$\begin{aligned} f'(x_1) &= f(x_0)\frac{2x_1 - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)} + f(x_1)\frac{2x_1 - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)} \\ &\quad + f(x_2)\frac{2x_1 - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)} + \prod_{i=0, i\neq 1}^{2}(x_1 - x_i)\frac{f'''(\xi(x_1))}{6} \\ &= -\frac{1}{2h}f(x_0) + \frac{1}{2h}f(x_2) - \frac{f'''(\xi(x_1))}{6}h^2 \end{aligned}$$

Derivation via Lagrange polynomial representation

- Pick $l = 1$:

$$\begin{aligned} f'(x_1) &= f(x_0)\frac{2x_1 - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)} + f(x_1)\frac{2x_1 - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)} \\ &\quad + f(x_2)\frac{2x_1 - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)} + \prod_{i=0, i\neq 1}^{2}(x_1 - x_i)\frac{f'''(\xi(x_1))}{6} \\ &= -\frac{1}{2h}f(x_0) + \frac{1}{2h}f(x_2) - \frac{f'''(\xi(x_1))}{6}h^2 \\ &= \frac{1}{2h}[f(x_1 + h) - f(x_1 - h)] - \frac{f'''(\xi(x_1))}{6}h^2. \end{aligned}$$

Derivation via Lagrange polynomial representation

- Pick $l = 1$:

$$\begin{aligned} f'(x_1) &= f(x_0)\frac{2x_1 - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)} + f(x_1)\frac{2x_1 - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)} \\ &\quad + f(x_2)\frac{2x_1 - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)} + \prod_{i=0, i\neq 1}^{2}(x_1 - x_i)\frac{f'''(\xi(x_1))}{6} \\ &= -\frac{1}{2h}f(x_0) + \frac{1}{2h}f(x_2) - \frac{f'''(\xi(x_1))}{6}h^2 \\ &= \frac{1}{2h}[f(x_1 + h) - f(x_1 - h)] - \frac{f'''(\xi(x_1))}{6}h^2. \end{aligned}$$

- Choose $x_1 = t_j$, then we get the centered difference:
  $f'(t_j) = \frac{1}{2h}[f(t_j + h) - f(t_j - h)] - \frac{f'''(\xi(t_j))}{6}h^2.$

Roundoff error

- Numerical differentiation: use the values of a function at mesh nodes to approximate the derivative values of the function.

Roundoff error

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- What if the values of a function at mesh nodes can not be expressed exactly due to roundoff?

Roundoff error

- Numerical differentiation: use the values of a function at mesh nodes to approximate the derivative values of the function.
- What if the values of a function at mesh nodes can not be expressed exactly due to roundoff?
- Conclusion: the total error caused by a small roundoff error could be HUGE!

Roundoff error

- Example 2: Consider the forward difference

$$f'(x_j) \approx \frac{f_{j+1} - f_j}{h} \left(= \frac{f(x_{j+1}) - f(x_j)}{h} \right).$$

Let $\tilde{f}_{j+1}$ and $\tilde{f}_j$ be the numbers stored in the computer memory for f_{j+1} and f_j respectively. Assume $f_{j+1} = \tilde{f}_{j+1} + e(x_{j+1})$ and $f_j = \tilde{f}_j + e(x_j)$ where $e(x_{j+1})$ and $e(x_j)$ are roundoff errors. In general $|e(x)| \leq \varepsilon |f(x)|$ for some relative error ε. Estimate the total error

$$\left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right|.$$

Roundoff error

- The forward difference scheme comes from the expansion

$$f'(x) = \frac{f(x+h) - f(x)}{h} + O(h)$$

Roundoff error

- The forward difference scheme comes from the expansion

$$f'(x) = \frac{f(x+h) - f(x)}{h} + O(h)$$

$$\Rightarrow \quad f'(x_j) = \frac{f(x_j+h) - f(x_j)}{h} + O(h)$$

Roundoff error

- The forward difference scheme comes from the expansion

$$f'(x) = \frac{f(x+h) - f(x)}{h} + O(h)$$

$$\Rightarrow \quad f'(x_j) = \frac{f(x_j+h) - f(x_j)}{h} + O(h)$$

$$\Rightarrow \quad f'(x_j) = \frac{f(x_{j+1}) - f(x_j)}{h} + O(h)$$

Roundoff error

- The forward difference scheme comes from the expansion

$$
\begin{aligned}
& f'(x) = \frac{f(x+h) - f(x)}{h} + O(h) \\
\Rightarrow \quad & f'(x_j) = \frac{f(x_j+h) - f(x_j)}{h} + O(h) \\
\Rightarrow \quad & f'(x_j) = \frac{f(x_{j+1}) - f(x_j)}{h} + O(h) \\
\Rightarrow \quad & f'(x_j) = \frac{f_{j+1} - f_j}{h} + O(h).
\end{aligned}
$$

Roundoff error

- Then

$$\left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right|$$

Roundoff error

- Then

$$\begin{aligned} & \left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\ \leq & \left| f'(x_j) - \frac{f_{j+1} - f_j}{h} \right| + \left| \frac{f_{j+1} - f_j}{h} - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \end{aligned}$$

Roundoff error

- Then

$$\begin{aligned} & \left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\ \leq & \left| f'(x_j) - \frac{f_{j+1} - f_j}{h} \right| + \left| \frac{f_{j+1} - f_j}{h} - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\ \leq & O(h) + \left| \frac{(f_{j+1} - \tilde{f}_{j+1}) - (f_j - \tilde{f}_j)}{h} \right| \end{aligned}$$

Roundoff error

- Then

$$
\begin{aligned}
& \left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\
\leq & \left| f'(x_j) - \frac{f_{j+1} - f_j}{h} \right| + \left| \frac{f_{j+1} - f_j}{h} - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\
\leq & O(h) + \left| \frac{(f_{j+1} - \tilde{f}_{j+1}) - (f_j - \tilde{f}_j)}{h} \right| \\
\leq & O(h) + \left| \frac{e(x_{j+1}) - e(x_j)}{h} \right|
\end{aligned}
$$

Roundoff error

- Then

$$
\begin{aligned}
& \left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\
\leq & \left| f'(x_j) - \frac{f_{j+1} - f_j}{h} \right| + \left| \frac{f_{j+1} - f_j}{h} - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\
\leq & O(h) + \left| \frac{(f_{j+1} - \tilde{f}_{j+1}) - (f_j - \tilde{f}_j)}{h} \right| \\
\leq & O(h) + \left| \frac{e(x_{j+1}) - e(x_j)}{h} \right| \\
\leq & O(h) + \frac{|e(x_{j+1})| + |e(x_j)|}{h}.
\end{aligned}
$$

Roundoff error

- Recall that $|e(x)| \leq \varepsilon |f(x)|$.
- Then

$$\left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right|$$

Roundoff error

- Recall that $|e(x)| \leq \varepsilon |f(x)|$.
- Then

$$\begin{aligned} & \left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\ \leq \quad & O(h) + \frac{|e(x_{j+1})| + |e(x_j)|}{h} \end{aligned}$$

Roundoff error

- Recall that $|e(x)| \leq \varepsilon |f(x)|$.
- Then

$$\begin{aligned} & \left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\ \leq \quad & O(h) + \frac{|e(x_{j+1})| + |e(x_j)|}{h} \\ \leq \quad & O(h) + \frac{\varepsilon |f(x_{j+1})| + \varepsilon |f(x_j)|}{h} \end{aligned}$$

Roundoff error

- Recall that $|e(x)| \leq \varepsilon |f(x)|$.
- Then

$$\begin{aligned} & \left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\ \leq \quad & O(h) + \frac{|e(x_{j+1})| + |e(x_j)|}{h} \\ \leq \quad & O(h) + \frac{\varepsilon |f(x_{j+1})| + \varepsilon |f(x_j)|}{h} \\ \leq \quad & O(h) + \frac{2\varepsilon}{h} \max_x |f(x)|. \end{aligned}$$

Roundoff error

- Recall that $|e(x)| \leq \varepsilon |f(x)|$.
- Then

$$\begin{aligned} & \left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\ \leq \quad & O(h) + \frac{|e(x_{j+1})| + |e(x_j)|}{h} \\ \leq \quad & O(h) + \frac{\varepsilon |f(x_{j+1})| + \varepsilon |f(x_j)|}{h} \\ \leq \quad & O(h) + \frac{2\varepsilon}{h} \max_x |f(x)|. \end{aligned}$$

- The error could be HUGE if h is much smaller than ε!

Roundoff error

- Example 2: Consider the forward difference for $f(x) = ln(x)$:

$$ln'(x) \approx \frac{ln(x+h) - ln(x)}{h}.$$

Compute the numerical differentiation for $x = 3$ and $h = 10^{-n}$ $(n = 1, 2, \cdots, 16)$. Use $ln'(x) = \frac{1}{x}$ to compute the errors $\left| ln'(3) - \frac{ln(3+h)-ln(3)}{h} \right|$ of the numerical solutions.

Roundoff error

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- Open your Matlab to code for this simple problem!

Roundoff error

- Numerical results:

h	numerical differentiation	error
10^{-1}	0.327898228229910	0.005435105103423
10^{-2}	0.332779009267470	0.000554324065863
10^{-3}	0.333277790120601	0.000055543212733
10^{-4}	0.333327777903847	0.000005555429486
10^{-5}	0.333332777802120	0.000000555531213
10^{-6}	0.333333277868775	0.000000055464559
10^{-7}	0.333333329827212	0.000000003506121
10^{-8}	0.333333338708997	0.000000000375663
10^{-9}	0.333333360913457	0.000000027580124
10^{-10}	0.333333360913457	0.000000027580124
10^{-11}	0.333333360913457	0.000000027580124
10^{-12}	0.333510996597397	0.000177663264064
10^{-13}	0.333066907387547	0.000266425945786
10^{-14}	0.355271367880050	0.021938034546717
10^{-15}	0.444089209850063	0.110755876516729
10^{-16}	0	0.333333333333333

Outline

1 Numerical differentiation

2 Numerical integration/quadrature

3 Other topics

Introduction

Basic idea:

- Consider $J(f) = \int_a^b f(x)\, dx$.
- Find numerical quadrature $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ to approximate $J(f)$.
- Analyze the error $E(f) = J(f) - Q(f)$.

Definition (quadrature weights and nodes)

In

$$Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j),$$

α_j $(j = 0, \cdots, m)$ are called quadrature weights and x_j $(j = 0, \cdots, m)$ are called quadrature nodes.

Introduction

- Example 1: Trapezoid rule

$$J(f) = \int_a^b f(x)\, dx \approx Q(f) = \frac{b-a}{2} f(a) + \frac{b-a}{2} f(b).$$

- Quadrature weights: $\alpha_0 = \alpha_1 = \frac{b-a}{2}$
- Quadrature nodes: $x_0 = a$ and $x_1 = b$.

Introduction

- Example 2: Simpson's rule

$$J(f) = \int_a^b f(x)\, dx$$
$$\approx \quad Q(f) = \frac{b-a}{6} f(a) + \frac{2(b-a)}{3} f(\frac{a+b}{2}) + \frac{b-a}{6} f(b).$$

- Quadrature weights: $\alpha_0 = \alpha_2 = \frac{b-a}{6}$, $\alpha_1 = \frac{2(b-a)}{3}$
- Quadrature nodes: $x_0 = a$, $x_1 = \frac{a+b}{2}$ and $x_2 = b$.
- Any observation for the quadrature weights?
- $\sum_{j=0}^{m} \alpha_j = b - a.$

Introduction

Remark

$\sum_{j=0}^{m} \alpha_j = b - a$ actually guarantees that $E(f) = 0$ for constant function $f(x) = c$:

$$\begin{aligned} E(f) &= J(f) - Q(f) = \int_a^b f(x)\, dx - \sum_{j=0}^{m} \alpha_j f(x_j) \\ &= \int_a^b c\, dx - \sum_{j=0}^{m} \alpha_j c = c(b-a) - c\sum_{j=0}^{m} \alpha_j \\ &= c(b-a) - c(b-a) \\ &= 0. \end{aligned}$$

That is, if $f(x)$ is a constant function, then the numerical quadrature is exact.

Introduction

Composite numerical integration:

- Divide $[a, b]$ into N pieces by using $N + 1$ points: $a_0 = a < a_1 < \cdots < a_{N-1} < a_N = b$.
- Then

$$J(f) = \int_a^b f(x)\, dx = \sum_{\nu=0}^{N-1} \int_{a_\nu}^{a_{\nu+1}} f(x)\, dx = \sum_{\nu=0}^{N-1} J_\nu(f).$$

- $J_\nu(f) \approx Q_\nu(f) = \sum_{j=0}^{m_\nu} \alpha_{\nu,j} f(x_{\nu,j})$.
- Hence

$$J(f) \approx Q(f) = \sum_{\nu=0}^{N-1} Q_\nu(f) = \sum_{\nu=0}^{N-1} \sum_{j=0}^{m_\nu} \alpha_{\nu,j} f(x_{\nu,j}).$$

Introduction

Affine mapping:

- Usually the numerical quadratures on the reference interval $[-1, 1]$ (or $[0, 1]$) are given. In practice, they need to be transferred to the numerical quadrature on arbitrary interval $[a, b]$ by affine mapping.
- If $x \in [a, b]$, then

$$a \le x \le b \Rightarrow -\frac{b-a}{2} \le x - \frac{a+b}{2} \le \frac{b-a}{2} \Rightarrow -1 \le \frac{x - \frac{a+b}{2}}{\frac{b-a}{2}} \le 1.$$

- Let $\hat{x} = \frac{x - \frac{a+b}{2}}{\frac{b-a}{2}}$. Then $\hat{x} \in [-1, 1]$ and $x = \frac{b-a}{2}\hat{x} + \frac{a+b}{2}$.

Introduction

- Hence

$$\begin{aligned} J(f) &= \int_a^b f(x)\, dx \\ &= \int_{-1}^1 f\left(\frac{b-a}{2}\hat{x} + \frac{a+b}{2}\right)\, d\left(\frac{b-a}{2}\hat{x} + a\right) \\ &= \int_{-1}^1 f\left(\frac{b-a}{2}\hat{x} + \frac{a+b}{2}\right)\frac{b-a}{2}\, d\hat{x}. \end{aligned}$$

- Assume we already have the numerical quadrature for $\hat{f}(\hat{x})$ on $[-1, 1]$:

$$\hat{J}(\hat{f}) = \int_{-1}^1 \hat{f}(\hat{x})\, d\hat{x} \approx \hat{Q}(\hat{f}) = \sum_{j=0}^m \hat{\alpha}_j \hat{f}(\hat{x}_j).$$

Introduction

- Let $\hat{f}(\hat{x}) = \frac{b-a}{2} f\left(\frac{b-a}{2}\hat{x} + \frac{a+b}{2}\right)$. Then

$$J(f) = \int_{-1}^{1} \frac{b-a}{2} f\left(\frac{b-a}{2}\hat{x} + \frac{a+b}{2}\right)\, d\hat{x} = \int_{-1}^{1} \hat{f}(\hat{x})\, d\hat{x}$$
$$\approx \quad \hat{Q}(\hat{f}) = \sum_{j=0}^{m} \hat{\alpha}_j \hat{f}(\hat{x}_j) = \sum_{j=0}^{m} \hat{\alpha}_j \frac{b-a}{2} f\left(\frac{b-a}{2}\hat{x}_j + \frac{a+b}{2}\right).$$

- Let $\alpha_j = \hat{\alpha}_j \frac{b-a}{2}$ and $x_j = \frac{b-a}{2}\hat{x}_j + \frac{a+b}{2}$. Then

$$J(f) = \int_a^b f(x)\, dx \approx Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j).$$

Introduction

- Homework: Assume the numerical quadrature for $\hat{f}(\hat{x})$ on $[0, 1]$ is

$$\hat{J}(\hat{f}) = \int_0^1 \hat{f}(\hat{x})\, d\hat{x} \approx \hat{Q}(\hat{f}) = \sum_{j=0}^{m} \hat{\alpha}_j \hat{f}(\hat{x}_j).$$

Derive the numerical quadrature of $J(f) = \int_a^b f(x)\, dx$.

Newton-Cotes formulas

Basic idea:

- The quadrature nodes x_j $(j = 0, \cdots, m)$ are taken to be equally spaced on $[a, b]$.
- Recall the *n*-th order approximation capability of the n-th order Lagrange polynomial approximation based on $(n+1)$-th nodes.
- Can we use the $m+1$ quadrature nodes to construct a quadrature rule which is exact for m-th order polynomial?
- Yes!

Newton-Cotes formulas

Recall

Theorem

For any $n+1$ distinct real numbers $x_0, x_1, x_2, \cdots, x_n$ and for $n+1$ arbitrary numbers $y_0, y_1, y_2, \cdots, y_n$, let

$$L_k(x) = \prod_{i=0, i\neq k}^{n} \frac{x - x_i}{x_k - x_i}.$$

Then there exists a unique interpolating polynomial

$$p(x) = \sum_{k=0}^{n} y_k L_k(x)$$

of degree at most n such that $y_j = p(x_j)$ $(j = 0, 1, 2, \cdots, n)$.

Newton-Cotes formulas

Recall

Definition (Lagrange polynomial approximation)

$p(x) = \sum_{k=0}^{n} f(x_k)L_k(x)$ is called the Lagrange polynomial approximation/interpolation of a given function $f(x)$.

Newton-Cotes formulas

Recall

Theorem

If $x_0, x_1, x_2, \cdots, x_n$ are $n+1$ distinct points in $[a, b]$ and $f \in C^{n+1}[a, b]$, then for each $x \in [a, b]$, there exists a number $\xi = \xi(x) \in (a, b)$ such that

$$f(x) - p(x) = R(x) = \frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!}$$

where $W(x) = \prod_{i=0}^{n}(x - x_i)$ and $p(x)$ is the Lagrange polynomial approximation/interpolation.

Newton-Cotes formulas

Construction of Newton-Cotes formulas:

- The quadrature nodes x_j $(j = 0, \cdots, m)$ are taken to be equally spaced on $[a, b]$.
- Assume $p(x)$ is the Lagrange polynomial approximation of $f(x)$ based on x_j $(j = 0, \cdots, m)$. Then

$$\begin{aligned} J(f) &= \int_a^b f(x)\, dx \\ &= \int_a^b p(x)\, dx + \int_a^b R(x)\, dx \\ &= \int_a^b \sum_{j=0}^{m} f(x_j) L_j(x)\, dx + \int_a^b R(x)\, dx \\ &= \sum_{j=0}^{m} f(x_j) \int_a^b L_j(x)\, dx + \int_a^b R(x)\, dx. \end{aligned}$$

Newton-Cotes formulas

- Let $\alpha_j = \int_a^b L_j(x)\ dx\ (j = 0, 1, \cdots, m)$. Then

$$J(f) \approx Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j).$$

- And

$$E(f) = J(f) - Q(f) = \int_a^b R(x)\ dx = \int_a^b \frac{f^{(m+1)}(\xi(x))W(x)}{(m+1)!}\ dx.$$

  where $W(x) = \prod_{j=0}^{m}(x - x_j)$.

- If f is a polynomial of degree $\leq m$, then $f^{(m+1)} \equiv 0 \Rightarrow E(f) = J(f) - Q(f) = 0$.

- Hence $Q(f)$ is exact at least for polynomials of degree $\leq m$. That is, $Q(f) \equiv J(f)$ when f is a polynomial of degree $\leq m$.

Newton-Cotes formulas

- Example 3: Compute the quadrature weights of the Newton-Cotes formula on [0,1] for $x_0 = 0$, $x_1 = \frac{1}{3}$, $x_2 = \frac{2}{3}$, $x_3 = 1$.
- First,

$$\begin{aligned}
\alpha_0 &= \int_0^1 L_0(x)\, dx = \int_0^1 \prod_{i=0, i\neq 0}^{m} \frac{x - x_i}{x_0 - x_i}\, dx \\
&= \int_0^1 \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)}\, dx \\
&= \int_0^1 \frac{(x - \frac{1}{3})(x - \frac{2}{3})(x - 1)}{(0 - \frac{1}{3})(0 - \frac{2}{3})(0 - 1)}\, dx \\
&= -\frac{9}{2} \int_0^1 \left(x^3 - 2x^2 + \frac{11}{9} - \frac{2}{9} \right)\, dx = \frac{1}{8}.
\end{aligned}$$

Newton-Cotes formulas

- Similarly, $\alpha_1 = \frac{3}{8}$, $\alpha_2 = \frac{3}{8}$, and $\alpha_3 = \frac{1}{8}$.
- Hence

$$\int_0^1 f(x)\, dx \approx \frac{1}{8}f(0) + \frac{3}{8}f(\frac{1}{3}) + \frac{3}{8}f(\frac{2}{3}) + \frac{1}{8}f(1).$$

Newton-Cotes formulas

Remark

There is another way to derive the Newton-Cotes formulas. The basic idea is to set up $m+1$ equations for α_j $(j = 0, 1, \cdots, m)$ by requiring $J(f) = Q(f)$ for $f = 1, x, x^2, \cdots, x^m$. (Independent study problem: see section 6.3.2.1 on page 336-338 of the textbook for more details).

Newton-Cotes formulas

Definition (closed Newton-Cotes formulas)

The $m+1$ point closed Newton-Cotes formulas have points $x_j = a + jh$ $(j = 0, 1, \cdots, m)$ where $h = \frac{b-a}{m}$.

Definition (open Newton-Cotes formulas)

The $m+1$ point open Newton-Cotes formulas have points $x_j = a + (j+1)h$ $(j = 0, 1, \cdots, m)$ where $h = \frac{b-a}{m+2}$.

Newton-Cotes formulas

- Example 4: two examples for closed and open Newton-Cotes formulas respectively.
- Closed Newton-Cotes formulas:
- Trapezoid rule: $m = 1$, $x_0 = a$, $x_1 = b$.
- First,

$$\begin{aligned}
\alpha_0 &= \int_a^b L_0(x)\, dx = \int_a^b \prod_{i=0, i\neq 0}^{1} \frac{x - x_i}{x_0 - x_i}\, dx \\
&= \int_a^b \frac{x-b}{a-b}\, dx = \frac{1}{a-b}\left(\frac{1}{2}x^2 - bx\right)\Big|_{x=a}^{x=b} \\
&= \frac{1}{a-b}\left[\left(\frac{1}{2}b^2 - b^2\right) - \left(\frac{1}{2}a^2 - ab\right)\right].
\end{aligned}$$

Newton-Cotes formulas

- Then

$$\begin{aligned}
\alpha_0 &= \frac{1}{a-b}\left[-\frac{1}{2}b^2 - \frac{1}{2}a^2 + ab\right] \\
&= \frac{1}{2(b-a)}(b^2 + a^2 - 2ab) \\
&= \frac{1}{2(b-a)}(b-a)^2 \\
&= \frac{b-a}{2}.
\end{aligned}$$

- Similarly, $\alpha_1 = \frac{b-a}{2}$.
- Hence the trapezoid rule:

$$J(f) = \int_a^b f(x)\,dx \approx Q(f) = \frac{b-a}{2}f(a) + \frac{b-a}{2}f(b).$$

Newton-Cotes formulas

- Simpson's rule (Homework): $m = 2$, $x_0 = a$, $x_1 = \frac{a+b}{2}$, $x_2 = b$.

$$J(f) = \int_a^b f(x)\, dx$$

$$\approx \quad Q(f) = \frac{b-a}{6} f(a) + \frac{2(b-a)}{3} f(\frac{a+b}{2}) + \frac{b-a}{6} f(b).$$

Newton-Cotes formulas

- Open Newton-Cotes formulas:
- Middle point rule (Homework): $m = 0$, $x_0 = \frac{a+b}{2}$.

$$J(f) = \int_a^b f(x)\, dx \approx Q(f) = (b-a) f(\frac{a+b}{2}).$$

- Two point open rule (Homeowrk): $m = 1$, $x_0 = a + \frac{b-a}{3}$, $x_1 = a + \frac{2(b-a)}{3}$.

$$\begin{aligned} J(f) &= \int_a^b f(x)\, dx \\ \approx \quad Q(f) &= \frac{b-a}{2} f(a + \frac{b-a}{3}) + \frac{b-a}{2} f(a + \frac{2(b-a)}{3}). \end{aligned}$$

Newton-Cotes formulas

- Example 5: use the Newton-Cotes formulas in Examples 3 and 4 to approximate $J(f) = \int_1^2 \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}\, dx$.
- Let $a = 1$, $b = 2$, $f(x) = \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}$. Then $J(f) = \int_a^b f(x)\, dx$.
- Trapezoid rule:

$$\begin{aligned} Q(f) &= \frac{b-a}{2}f(a) + \frac{b-a}{2}f(b) \\ &= \frac{1}{2}\frac{cos\left(\frac{\pi}{4}\right)}{sin^2\left(\frac{\pi}{4}\right)} + \frac{1}{2}\frac{cos\left(\frac{\pi}{2}\right)}{sin^2\left(\frac{\pi}{2}\right)} \\ &= \frac{1}{2}\cdot\frac{\frac{\sqrt{2}}{2}}{\frac{1}{2}} + \frac{1}{2}\cdot\frac{0}{1} \\ &= \frac{\sqrt{2}}{2} \approx 0.707106781186548. \end{aligned}$$

Newton-Cotes formulas

- Analytic solution: substitution rule with $u = sin\left(\frac{\pi}{4}x\right)$.

$$\begin{aligned}\int \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}\,dx &= \frac{4}{\pi}\int \frac{1}{u^2}\,du = -\frac{4}{\pi}\frac{1}{u} \\ &= -\frac{4}{\pi}\frac{1}{sin\left(\frac{\pi}{4}x\right)}.\end{aligned}$$

- Hence

$$\begin{aligned}J(f) &= \int_1^2 \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}\,dx = -\frac{4}{\pi}\frac{1}{sin\left(\frac{\pi}{4}x\right)}\Big|_{x=1}^{x=2} \\ &= -\frac{4}{\pi}\frac{1}{sin\left(\frac{\pi}{2}\right)} + \frac{4}{\pi}\frac{1}{sin\left(\frac{\pi}{4}\right)} \\ &= -\frac{4}{\pi} + \frac{4}{\pi}\sqrt{2} \approx 0.527393087579050.\end{aligned}$$

Newton-Cotes formulas

- The error of trapezoid rule is

$$\begin{aligned} |Q(f) - J(f)| &\approx |0.707106781186548 - 0.527393087579050| \\ &= 0.179713693607498. \end{aligned}$$

- Homework: use the other Newton-Cotes formulas in Examples 3 and 4 to approximate $J(f)$ by hand.
- What can we do if we have thousands of or even millions of integrations to compute?
- Computer program!
- Open your matlab to code for this problem!
- One observation: all numerical quadratures are $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ with different quadrature weights and nodes.

Newton-Cotes formulas

- Numerical results:

Newton-Cotes formulas	numerical integration	error
Middle point rule	0.448341529167965	0.079051558411084
Two point open rule	0.472034041575363	0.055359046003686
Trapezoid rule	0.707106781186548	0.179713693607498
Simpson's rule	0.534596613174159	0.007203525595110
Four point closed rule		

Newton-Cotes formulas

- For the four point closed rule on $[0, 1]$ in Example 3, we can use affine mapping to obtain the quadrature weights and nodes on $[a, b]$.
- Assume $\hat{Q}(\hat{f}) = \sum_{j=0}^{m} \hat{\alpha}_j \hat{f}(\hat{x}_j)$ is the quadrature formula on $[0, 1]$ and $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ is the corresponding quadrature formula on $[a, b]$.
- Then by the homework of affine mapping, we have $\alpha_j = \hat{\alpha}_j(b - a)$ and $x_j = (b - a)\hat{x}_j + a$.

Newton-Cotes formulas

- Numerical results:

Newton-Cotes formulas	numerical integration	error
Middle point rule	0.448341529167965	0.079051558411084
Two point open rule	0.472034041575363	0.055359046003686
Trapezoid rule	0.707106781186548	0.179713693607498
Simpson's rule	0.534596613174159	0.007203525595110
Four point closed rule	0.530802226478159	0.003409138899110

Newton-Cotes formulas

- Now we analyze the error $E(f)$ in detail.
- Recall that

$$\begin{aligned} E(f) &= \int_a^b \frac{f^{(m+1)}(\xi(x))W(x)}{(m+1)!}\,dx \\ &= \int_a^b \prod_{j=0}^{m}(x-x_j)\frac{f^{(m+1)}(\xi(x))}{(m+1)!}\,dx. \end{aligned}$$

- Then

$$\begin{aligned} |E(f)| &\leq \int_a^b \left|\prod_{j=0}^{m}(x-x_j)\right| \left|\frac{f^{(m+1)}(\xi(x))}{(m+1)!}\right| dx \\ &\leq \frac{\left\|f^{(m+1)}\right\|_\infty}{(m+1)!}\int_a^b \left|\prod_{j=0}^{m}(x-x_j)\right| dx. \end{aligned}$$

Newton-Cotes formulas

- Consider the following change of variables:

$$z = \frac{x-a}{b-a}, z_i = \frac{x_i - a}{b-a}.$$

- Then $x - x_i = (z - z_i)(b-a)$ and $dx = (b-a)dz$.
- Hence

$$|E(f)| \leq \frac{\left\|f^{(m+1)}\right\|_\infty}{(m+1)!}(b-a)^{m+2}\int_0^1 \left|\prod_{j=0}^{m}(z-z_j)\right| dz.$$

- Let $H = b - a$ and $\beta^* = \int_0^1 \left|\prod_{j=0}^m (z - z_j)\right| dz$. Then

$$|E(f)| \leq H^{m+2}\beta^* \frac{\left\|f^{(m+1)}\right\|_\infty}{(m+1)!}.$$

Newton-Cotes formulas

Remark

The Newton-Cotes formula with $m+1$ nodes have accuracy order of at least $m+2$.

Newton-Cotes formulas

- Example 6: Analyze the error of trapezoid rule: $m = 1$, $x_0 = a$, $x_1 = b$.

$$J(f) = \int_a^b f(x)\, dx \approx Q(f) = \frac{b-a}{2} f(a) + \frac{b-a}{2} f(b).$$

- Recall that

$$E(f) = \int_a^b \prod_{j=0}^{m} (x - x_j) \frac{f^{(m+1)}(\xi(x))}{(m+1)!}\, dx.$$

Newton-Cotes formulas

- Hence for the trapezoid rule, we have

$$\begin{aligned}
|E(f)| &= \left| \int_a^b (x-a)(x-b) \frac{f''(\xi(x))}{2} \, dx \right| \\
&\leq \frac{\|f''\|_\infty}{2} \int_a^b |(x-a)(x-b)| \, dx \\
&= -\frac{\|f''\|_\infty}{2} \int_a^b (x-a)(x-b) \, dx \\
&= \frac{\|f''\|_\infty}{12} (b-a)^3 \\
&= \frac{\|f''\|_\infty}{12} H^3.
\end{aligned}$$

Newton-Cotes formulas

- Example 7: Analyze the error of middle point rule: $m = 0$, $x_0 = \frac{a+b}{2}$.

$$J(f) = \int_a^b f(x)\, dx \approx Q(f) = (b-a) f(\frac{a+b}{2}).$$

- It's expected that the error is second order since $m = 0$.
- However, by using Taylor expansion, a third order accuracy can be shown for the middle point rule.
- First, by Taylor expansion, we get

$$f(x) = f(\frac{a+b}{2}) + f'(\frac{a+b}{2})(x - \frac{a+b}{2}) + \frac{1}{2} f''(\xi)(x - \frac{a+b}{2})^2.$$

Newton-Cotes formulas

- Then

$$\int_a^b f(x)\,dx = \int_a^b f(\frac{a+b}{2})\,dx + \int_a^b f'(\frac{a+b}{2})(x-\frac{a+b}{2})\,dx + \int_a^b \frac{1}{2}f''(\xi)(x-\frac{a+b}{2})^2\,dx.$$

- Note that

$$\begin{aligned}\int_a^b (x-\frac{a+b}{2})\,dx &= \left[\frac{1}{2}x^2 - \frac{a+b}{2}x\right]\Big|_{x=a}^{x=b} \\ &= \left[\frac{1}{2}b^2 - \frac{a+b}{2}b\right] - \left[\frac{1}{2}a^2 - \frac{a+b}{2}a\right] \\ &= \left(-\frac{1}{2}ab\right) - \left(-\frac{1}{2}ab\right) \\ &= 0.\end{aligned}$$

Newton-Cotes formulas

- Similarly,

$$\int_a^b (x - \frac{a+b}{2})^2 \, dx = \frac{1}{12}(b-a)^3.$$

- Hence

$$\begin{aligned}\int_a^b f(x) \, dx &= \int_a^b f(\frac{a+b}{2}) \, dx + \int_a^b f'(\frac{a+b}{2})(x - \frac{a+b}{2}) \, dx \\ &\quad + \int_a^b \frac{1}{2} f''(\xi)(x - \frac{a+b}{2})^2 \, dx \\ &= (b-a) f(\frac{a+b}{2}) + \frac{1}{24}(b-a)^3 f''(\xi).\end{aligned}$$

Newton-Cotes formulas

Remark

If m is odd, the Newton-Cotes formula with $m+1$ nodes has accuracy order of $m+2$ and is exact for polynomials of degree m. If m is even, the Newton-Cotes formula with $m+1$ nodes has accuracy order of $m+3$ and is exact for polynomials of degree $m+1$. (Independent study problem: see Remark 6.4 on page 341-342 of the textbook for more details.)

Remark

There is another way to derive the Newton-Cotes formula, which is to choose the quadrature weights α_j such that the quadrature formula $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ is exact for polynomials of degree $\leq m$. (Independent study problem: see section 6.3.2.1 on page 336-338 for more details.)

Gaussian quadrature

Basic idea:

- Use $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ to approximate $J(f) = \int_a^b f(x)\, dx$.
- Newton-Cotes formulas: find the quadrature weights α_j with fixed quadrature nodes such that the quadrature formula is exact ($Q(f) \equiv J(f)$) for the highest degree polynomial possible.
- Gaussian quadrature: find the quadrature weights α_j and nodes x_j such that the quadrature formula is exact for the highest degree polynomial possible.
- How many equations are needed to uniquely determine α_j and x_j ($j = 0, \cdots, m$)? 2(m+1).
- How to set up the $2(m+1)$ equations?
- Require the quadrature formula to be exact for $f(x) = 1, x, x^2, \cdots, x^{2m+1}$.

Gaussian quadrature

- Example 8: Find the Gaussian quadrature of two nodes ($m = 1$) for $\int_{-1}^{1} f(x)\ dx$.
- $\int_{-1}^{1} f(x)\ dx \approx Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j) = \alpha_0 f(x_0) + \alpha_1 f(x_1)$.
- $m = 1 \Rightarrow 2m + 1 = 3$.
- Require this quadrature formula to be exact for $f(x) = 1, x, x^2, x^3$.
- $\alpha_0 f(x_0) + \alpha_1 f(x_1) = \int_{-1}^{1} f(x)\ dx$ for $f(x) = 1, x, x^2, x^3$.
- For $f(x) = 1$: $\alpha_0 + \alpha_1 = \int_{-1}^{1} 1\ dx = 2$.
- For $f(x) = x$: $\alpha_0 x_0 + \alpha_1 x_1 = \int_{-1}^{1} x\ dx = 0$.
- For $f(x) = x^2$: $\alpha_0 x_0^2 + \alpha_1 x_1^2 = \int_{-1}^{1} x^2\ dx = \frac{2}{3}$.
- For $f(x) = x^3$: $\alpha_0 x_0^3 + \alpha_1 x_1^3 = \int_{-1}^{1} x^3\ dx = 0$.

Gaussian quadrature

- Hence

$$\begin{aligned}
\alpha_0 + \alpha_1 &= 2,\\
\alpha_0 x_0 + \alpha_1 x_1 &= 0,\\
\alpha_0 x_0^2 + \alpha_1 x_1^2 &= \frac{2}{3},\\
\alpha_0 x_0^3 + \alpha_1 x_1^3 &= 0.
\end{aligned}$$

  There are different techniques to solve this nonlinear system.

- If $\alpha_1 = 0$, then there is no solution for the system.
- If $x_0 = 0$, then there is no solution for the system.
- $\alpha_0 x_0 + \alpha_1 x_1 = 0 \Rightarrow \alpha_0 x_0 = -\alpha_1 x_1 \Rightarrow \frac{\alpha_0}{\alpha_1} = -\frac{x_1}{x_0}$.
- $\alpha_0 x_0^3 + \alpha_1 x_1^3 = 0 \Rightarrow \alpha_0 x_0^3 = -\alpha_1 x_1^3 \Rightarrow \frac{\alpha_0}{\alpha_1} = -\left(\frac{x_1}{x_0}\right)^3$.
- Then $-\frac{x_1}{x_0} = -\left(\frac{x_1}{x_0}\right)^3 \Rightarrow \frac{x_1}{x_0} = 0, 1, -1$.

Gaussian quadrature

- If $\frac{x_1}{x_0} = 0 \Rightarrow x_1 = 0$, then there is no solution for the system.
- If $\frac{x_1}{x_0} = 1 \Rightarrow x_1 = x_0$, then there is no solution for the system.
- Then $\frac{x_1}{x_0} = -1 \Rightarrow x_1 = -x_0$.
- Hence the system becomes

$$\begin{aligned} \alpha_0 + \alpha_1 &= 2, \\ \alpha_0 x_0 - \alpha_1 x_0 &= 0, \\ \alpha_0 x_0^2 + \alpha_1 x_0^2 &= \frac{2}{3}, \\ \alpha_0 x_0^3 - \alpha_1 x_0^3 &= 0. \end{aligned}$$

- Since $x_0 \neq 0$, then

$$\alpha_0 + \alpha_1 = 2, \ \alpha_0 - \alpha_1 = 0, \ \alpha_0 x_0^2 + \alpha_1 x_0^2 = \frac{2}{3}.$$

Gaussian quadrature

- $\alpha_0 + \alpha_1 = 2,\ \alpha_0 - \alpha_1 = 0 \Rightarrow \alpha_0 = 1,\ \alpha_1 = 1.$
- Then $\alpha_0 x_0^2 + \alpha_1 x_0^2 = \frac{2}{3} \Rightarrow 2x_0^2 = \frac{2}{3} \Rightarrow x_0 = \pm\frac{1}{\sqrt{3}}.$
- Then $x_1 = -x_0 = \mp\frac{1}{\sqrt{3}}.$
- Therefore, $\alpha_0 = 1,\ \alpha_1 = 1,\ x_0 = -\frac{1}{\sqrt{3}},\ x_1 = \frac{1}{\sqrt{3}}.$
- Finally,

$$\int_{-1}^{1} f(x)\ dx \approx Q(f) = \alpha_0 f(x_0) + \alpha_1 f(x_1) = f(-\frac{1}{\sqrt{3}}) + f(\frac{1}{\sqrt{3}}).$$

Gaussian quadrature

Remark

The matlab command "solve" may be used to obtain the analytic solution of a simple system of equations.

- Homework: Find the Gaussian quadrature of three nodes ($m = 2$) for $\int_{-1}^{1} f(x)\, dx$ by using the technique in Example 8.

Gaussian quadrature

- Question: Is there a general way to construct the Gauss quadrature?
- Yes, orthogonal polynomials!
- Step 1: choose the roots of the orthogonal polynomials to be the quadrature nodes.
- Step 2: use Lagrange polynomial approximation to obtain the quadrature weights. This is the same as Newton-Cotes formula.
- Consider $J(f) = \int_a^b f(x)\rho(x)\,dx$ where $\rho(x)$ is a real, positive, piecewise continuous weight function on (a, b).

Gaussian quadrature

Definition (Gaussian quadrature)

$Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ is called a Gaussian quadrature formula if

$$\int_a^b x^k \rho(x)\, dx = \sum_{j=0}^{m} \alpha_j x_j^k \quad \left(J(x^k) = Q(x^k) \right)$$

for $k = 0, 1, 2, \cdots, 2m+1$. That is, the quadrature formula is exact for polynomials of degree $\leq 2m+1$.

Gaussian quadrature

Definition

The inner product of $f(x)$ and $g(x)$ with respect to $\rho(x)$ is defined by

$$(f, g) = \int_a^b f(x)g(x)\rho(x)\ dx.$$

Definition (Orthogonal functions)

Functions $f(x)$ and $g(x)$ are orthogonal to each other if $(f, g) = 0$.

Gaussian quadrature

Definition (Orthogonal polynomials)

The set of polynomials $\{p_i(x)\}_{i=0}^{\infty}$ is called orthogonal on $[a, b]$ with respect to $\rho(x)$ if $p_i(x)$ $(i = 0, 1, \cdots)$ are orthogonal mutually, that is,

$$\int_a^b p_i(x)p_j(x)\rho(x)\ dx = 0\ \ ((p_i, p_j) = 0)$$

for $i \neq j$.

Gaussian quadrature

Remark

Recall the Gram-Schmidt orthogonalization process (Section 4.2) to find the set $\{p_i(x)\}_{i=0}^{\infty}$ *of orthogonal polynomials as follows:*

$$p_0(x) = 1,$$

$$p_i(x) = x^i - \sum_{k=0}^{i-1} \frac{1}{(p_k, p_k)}(x^j, p_k)p_k(x), \ \ i = 1, 2, \cdots .$$

Gaussian quadrature

Recall

Theorem

For any $n+1$ distinct real numbers $x_0, x_1, x_2, \cdots, x_n$ and for $n+1$ arbitrary numbers $y_0, y_1, y_2, \cdots, y_n$, let

$$L_k(x) = \prod_{i=0, i\neq k}^{n} \frac{x - x_i}{x_k - x_i}.$$

Then there exists a unique interpolating polynomial

$$p(x) = \sum_{k=0}^{n} y_k L_k(x)$$

of degree at most n such that $y_j = p(x_j)$ $(j = 0, 1, 2, \cdots, n)$.

Gaussian quadrature

Recall

Definition (Lagrange polynomial approximation)

$p(x) = \sum_{k=0}^{n} f(x_k) L_k(x)$ is called the Lagrange polynomial approximation/interpolation of a given function $f(x)$.

Gaussian quadrature

Recall

Theorem

If $x_0, x_1, x_2, \cdots, x_n$ are $n+1$ distinct points in $[a, b]$ and $f \in C^{n+1}[a, b]$, then for each $x \in [a, b]$, there exists a number $\xi = \xi(x) \in (a, b)$ such that

$$f(x) - p(x) = R(x) = \frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!}$$

where $W(x) = \prod_{i=0}^{n}(x - x_i)$ and $p(x)$ is the Lagrange polynomial approximation/interpolation.

Gaussian quadrature

Construction of Gaussian quadrature:

- The quadrature nodes x_j $(j = 0, \cdots, m)$ are taken to be the $m+1$ distinct roots of p_{m+1} in (a, b).
- Assume $p(x)$ is the Lagrange polynomial approximation of $f(x)$ based on x_j $(j = 0, \cdots, m)$. Then

$$\begin{aligned} J(f) &= \int_a^b f(x)\rho(x)\, dx \\ &= \int_a^b p(x)\rho(x)\, dx + \int_a^b R(x)\rho(x)\, dx \\ &= \int_a^b \sum_{j=0}^{m} f(x_j)L_j(x)\rho(x)\, dx + \int_a^b R(x)\rho(x)\, dx \\ &= \sum_{j=0}^{m} f(x_j)\int_a^b L_j(x)\rho(x)\, dx + \int_a^b R(x)\rho(x)\, dx. \end{aligned}$$

Gaussian quadrature

- Let $\alpha_j = \int_a^b L_j(x)\rho(x)\ dx\ (j = 0, 1, \cdots, m)$. Then

$$J(f) \approx Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j).$$

- And

$$E(f) = J(f) - Q(f) = \int_a^b R(x)\rho(x)\ dx = \int_a^b \frac{f^{(m+1)}(\xi(x))W(x)}{(m+1)!}\rho(x)\ dx.$$

  where $W(x) = \prod_{j=0}^{m}(x - x_j)$.

- If f is a polynomial of degree $\leq m$, then $f^{(m+1)} \equiv 0 \Rightarrow E(f) = J(f) - Q(f) = 0$.

- Hence $Q(f)$ is exact at least for polynomials of degree $\leq m$.

Gaussian quadrature

Theorem (I)

Consider the quadrature formula $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ *for* $J(f) = \int_a^b f(x)\rho(x)\,dx$. $Q(f)$ *is exact for polynomials of degree* $\leq 2m+1$ *if and only if the quadrature nodes* x_j $(j = 0, 1, \cdots, m)$ *are the roots of* $p_{m+1}(x)$ *and the quadrature weights* $\alpha_j = \int_a^b L_j(x)\rho(x)\,dx$ $(j = 0, 1, \cdots, m)$.

Proof.

See pages ?-? of the textbook. (Independent study problem or take Math 6601 (Numerical Analysis)) □

Gaussian quadrature

Remark

Except the Gram-Schmidt process, there is another algebraic procedure to obtain $p_{m+1}(x)$. Then the quadrature nodes and weights can be obtained by using Theorem (I). (Independent study problem: see Remark 6.11 on page 347-348 of the textbook for more details.)

Gaussian quadrature

Theorem (II)

Let $\rho(x) > 0$ be a weight function defined on $[a, b]$ and let $\{p_k(x)\}_{k=0}^{\infty}$ be the associated sequence of orthogonal polynomials generated by the Gram-Schmidt orthogonalization process. Let $\{x_i\}_{i=0}^{m}$ and $\{\alpha_i\}_{i=0}^{m}$ be the points and weights for Gauss quadratures respectively. If $f \in C^{2m+2}[a, b]$, we have

$$\int_a^b \rho(x) f(x)\, dx - (b-a) \sum_{j=0}^{m} \alpha_j f(x_j) = \frac{f^{(2m+2)}(\xi)}{(2m+2)!} \int_a^b \rho(x) p_{m+1}^2(x)\, dx$$

where $\xi \in (a, b)$.

Gaussian quadrature

Proof.

See page 349-350 of the textbook. (Independent study problem) □

Remark

The regularity requirement $f \in C^{2m+2}[a, b]$ is critical in practice!

Gaussian quadrature

Theorem (III)

The weights α_j $(j = 0, 1, \cdots, m)$ in Gaussian quadrature are all positive for any m.

Proof.

See page 352 of the textbook. (Independent study problem) □

Gaussian quadrature

Recall affine mapping:

- If $x \in [a, b]$, then

$$a \le x \le b \Rightarrow -\frac{b-a}{2} \le x - \frac{a+b}{2} \le \frac{b-a}{2} \Rightarrow -1 \le \frac{x - \frac{a+b}{2}}{\frac{b-a}{2}} \le 1.$$

- Let $\hat{x} = \frac{x - \frac{a+b}{2}}{\frac{b-a}{2}}$. Then $\hat{x} \in [-1, 1]$ and $x = \frac{b-a}{2}\hat{x} + \frac{a+b}{2}$.
- Then

$$\begin{aligned} J(f) &= \int_a^b f(x)\, dx = \int_{-1}^1 f\left(\frac{b-a}{2}\hat{x} + \frac{a+b}{2}\right)\, d\left(\frac{b-a}{2}\hat{x} + a\right) \\ &= \int_{-1}^1 f\left(\frac{b-a}{2}\hat{x} + \frac{a+b}{2}\right)\frac{b-a}{2}\, d\hat{x}. \end{aligned}$$

- Thus we only need to consider the Gauss quadrature on $[-1, 1]$.

Gaussian quadrature

- Example 9: Gauss-Legendre quadrature rules on $[-1, 1]$, which use $\rho(x) = 1$.
- m=0: $x_0 = 0$; $\alpha_0 = 2$. (middle point rule)
- m=1: $x_0 = -\frac{1}{\sqrt{3}}$, $x_1 = -x_0 = \frac{1}{\sqrt{3}}$; $\alpha_0 = \alpha_1 = 1$. (Example 8)
- m=2: $x_0 = -\sqrt{\frac{3}{5}}$, $x_1 = 0$, $x_2 = -x_0 = \sqrt{\frac{3}{5}}$; $\alpha_0 = \alpha_2 = \frac{5}{9}$, $\alpha_1 = \frac{8}{9}$.
- m=3: $x_0 = -\sqrt{\frac{3+\sqrt{4.8}}{7}}$, $x_1 = -\sqrt{\frac{3-\sqrt{4.8}}{7}}$, $x_2 = -x_1 = \sqrt{\frac{3-\sqrt{4.8}}{7}}$, $x_3 = -x_0 = \sqrt{\frac{3+\sqrt{4.8}}{7}}$; $\alpha_0 = \alpha_3 = \frac{1}{2} - \frac{1}{3\sqrt{4.8}}$, $\alpha_1 = \alpha_2 = \frac{1}{2} + \frac{1}{3\sqrt{4.8}}$.

Gaussian quadrature

- Let's derive the Gauss-Legendre quadrature rule on $[-1, 1]$ with $m = 1$.
- Recall

Theorem (I)

Consider the quadrature formula $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ *for* $J(f) = \int_a^b f(x)\rho(x)\,dx$. $Q(f)$ *is exact for polynomials of degree* $\leq 2m+1$ *if and only if the quadrature nodes* x_j $(j = 0, 1, \cdots, m)$ *are the roots of* $p_{m+1}(x)$ *and the quadrature weights* $\alpha_j = \int_a^b L_j(x)\rho(x)\,dx$ $(j = 0, 1, \cdots, m)$.

- Since $m = 1$, we first need to obtain the orthogonal polynomials $p_0(x), p_1(x), p_2(x)$ with respect to the weight function $\rho(x) = 1$.

Gaussian quadrature

- Gram-Schmidt process for $u_1 = 1, u_2 = x, u_3 = x^2$ with respect to the weight function $\rho(x) = 1$:

$$
\begin{aligned}
v_1 &= u_1 = 1, \\
\|v_1\|^2 &= (v_1, v_1) = \int_{-1}^{1} 1 \, dx = 2, \\
v_2 &= u_2 - \frac{(u_2, v_1)}{(v_1, v_1)} v_1 = x - \frac{1}{2} \int_{-1}^{1} x \, dx = x, \\
\|v_2\|^2 &= (v_2, v_2) = \int_{-1}^{1} x^2 \, dx = 2/3, \\
v_3 &= u_3 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2 \\
&= x^2 - \frac{1}{2} \int_{-1}^{1} x^2 \, dx - \frac{x}{2/3} \int_{-1}^{1} x^3 \, dx \\
&= x^2 - \frac{1}{2} \cdot \frac{2}{3} - \frac{x}{2/3} \cdot 0 = x^2 - \frac{1}{3}.
\end{aligned}
$$

Gaussian quadrature

- $p_2(x) = v_3 = x^2 - \frac{1}{3}$.
- So $x_0 = -\frac{\sqrt{3}}{3}$, $x_1 = \frac{\sqrt{3}}{3}$.
- Then

$$\begin{aligned}
\alpha_0 &= \int_{-1}^{1} L_0(x)\rho(x)\, dx = \int_{-1}^{1} \frac{x - x_1}{x_0 - x_1}\, dx \\
&= \int_{-1}^{1} \frac{x - \frac{\sqrt{3}}{3}}{-\frac{\sqrt{3}}{3} - \frac{\sqrt{3}}{3}}\, dx = \left(-\frac{\sqrt{3}}{4}x^2 + \frac{1}{2}x \right) \Big|_{x=-1}^{x=1} \\
&= 1.
\end{aligned}$$

- Similarly, $\alpha_1 = 1$.
- Hence, for $m = 1$: $x_0 = -\frac{1}{\sqrt{3}}$, $x_1 = -x_0 = \frac{1}{\sqrt{3}}$; $\alpha_0 = \alpha_1 = 1$.

Gaussian quadrature

- Example 10: use the Gauss formulas in Example 9 to approximate $J(f) = \int_1^2 \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}\, dx$, which is the same integral as in Example 5.
- Let $a = 1$, $b = 2$, $f(x) = \frac{cos\left(\frac{\pi}{4}x\right)}{sin^2\left(\frac{\pi}{4}x\right)}$. Then $J(f) = \int_a^b f(x)\, dx$.
- Recall the analytic solution from Example 5:

$$J(f) = -\frac{4}{\pi} + \frac{4}{\pi}\sqrt{2} \approx 0.527393087579050.$$

Practice in Matlab for Gaussian quadrature

- Recall the numerical results of Example 5:

Newton-Cotes formulas	numerical integration	error
Middle point rule	0.448341529167965	0.079051558411084
Two point open rule	0.472034041575363	0.055359046003686
Trapezoid rule	0.707106781186548	0.179713693607498
Simpson's rule	0.534596613174159	0.007203525595110
Four point closed rule	0.530802226478159	0.003409138899110

Practice in Matlab for Gaussian quadrature

- For the Gaussian quadratures on $[-1, 1]$ in Example 9, we can use affine mapping to obtain the quadrature weights and nodes on $[a, b]$.
- Assume $\hat{Q}(\hat{f}) = \sum_{j=0}^{m} \hat{\alpha}_j \hat{f}(\hat{x}_j)$ is the quadrature formula on $[-1, 1]$ and $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ is the corresponding quadrature formula on $[a, b]$.
- Then by the lecture of affine mapping, we have $\alpha_j = \hat{\alpha}_j \frac{b-a}{2}$ and $x_j = \frac{b-a}{2} \hat{x}_j + \frac{a+b}{2}$.

Practice in Matlab for Gaussian quadrature

- Numerical results:

Newton-Cotes formulas	numerical integration	error
Middle point rule	0.448341529167965	0.079051558411084
Two point open rule	0.472034041575363	0.055359046003686
Trapezoid rule	0.707106781186548	0.179713693607498
Simpson's rule	0.534596613174159	0.007203525595110
Four point closed rule	0.530802226478159	0.003409138899110
One point Gaussian	0.448341529167965	0.079051558411084
Two point Gaussian	0.522795202310018	0.004597885269032
Three point Gaussian	0.527191770561494	0.000201317017555
Four point Gaussian	0.527385259852249	0.000007827726801

Outline

1 Numerical differentiation

2 Numerical integration/quadrature

3 Other topics

Topics in numerical quadrature: Independent study problems

- Analysis of Gauss quadratures (Section 6.3.3.2): error analysis, positiveness of the Gaussian quadrature weights, etc.
- Romberg integration (Section 6.3.4): use extrapolation to improve the accuracy of a composite numerical integration.
- Multi-integrals, singular-Integrals, infinity integrals (Section 6.3.5): tensor product, sparse grid, truncation, change of variables, etc.
- Monte Carlo methods (Section 6.3.6): sampling.

Topics in numerical quadrature: Independent study problems

- Adaptive quadratures (Section 6.3.7): for a composite numerical integration, the quadrature itself should be able to detect which portions of the interval of integration need to be small and which portions can be large.

- Interval bounds (Section 6.3.8): Mathematically rigorous bounds on integrals can be computed. That is, a small interval in which the integral lies can be found.

Topics in numerical differentiation: Independent study problems

- Higher order difference schemes for $f'(x)$ by using more nodes.
- Finite difference schemes for higher order derivatives $f^{(n)}(x)$.
- Improve the accuracy order by choosing different set of nodes but keeping the same number of nodes.
- Automatic differentiation (Section 6.2)
- .....